PS3

DIFFUSION EQUATION

$$\begin{cases} \vec{J} = -D\nabla c \\ \dfrac{\partial c}{\partial t} = -\nabla\cdot\vec{J} \end{cases}$$

$$0 = D\nabla^2 c$$

$$\uparrow$$

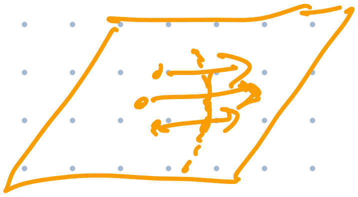
$$= D\left(\frac{1}{r}\frac{\partial}{\partial r} r \frac{\partial c}{\partial r}\right)$$

. . .

i) $$\boxed{c(r) \overset{?}{=} \frac{R_0 J_0}{D}\ln\left(\frac{R_0}{r}\right) + c_0}$$

IN 2d $[c]$ - PARTICLES/μm^2

$[J_0]$ - PARTICLES/$s\cdot\mu m$



D - $\dfrac{\mu m^2}{s}$

$$\left[\frac{R_0 J_0}{D}\right] - \frac{\cancel{\mu m}}{\mu m^2}\cdot\frac{\text{PARTICLES}}{\cancel{s}\cdot\cancel{\mu m}}\cdot\cancel{s}$$

ASIDE $$0 = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial c}{\partial r}\right)$$

$$0 = \frac{\partial}{\partial r}\left(r\frac{\partial c}{\partial r}\right)$$

$$r\frac{\partial c}{\partial r} = A$$

$$\frac{\partial c}{\partial r} = \frac{A}{r}$$

$$c(r) = A\ln\frac{\cancel{A}}{r} + B$$

ii)

$$\frac{\partial c}{\partial t} = D\nabla^2 c - \tau$$

$$0 = D\left(\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial c}{\partial r}\right)\right) - \tau$$

$$\dots \Rightarrow \frac{\partial}{\partial r}\left(r\frac{\partial c}{\partial r}\right) = \frac{\tau r}{D}$$

$$r\frac{\partial c}{\partial r} = \frac{1}{2}\frac{\tau r^2}{D} + \overset{\text{CONST}}{\underline{C_1}}$$

?

$$\Rightarrow c(r) = \frac{\gamma r^2}{4D} + \overset{\downarrow}{\underline{C_1}} \ln r + \overset{\downarrow}{\underline{C_2}}$$

✓✓ ✓✓

? ✓

$$\nparallel$$

$$\frac{\gamma (r^2 - R_0^2)}{4D}$$